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Self-Adjusting Tree (Splay Tree)

- ✓ A Self-adjusting Tree is a special type of Binary Search Tree (BST) that automatically changes its structure after operations like searching, insertion, and deletion. The most common self-adjusting tree is the Splay Tree, introduced by Sleator and Tarjan.
- ✓ The main idea of a splay tree is that whenever a node is accessed, that node is moved closer to the root by using special rotations called splaying. This helps frequently used elements become faster to access later.
- ✓ Unlike AVL trees or Red-Black trees, a splay tree does not keep strict balance information such as balance factors or colors. Instead, it depends only on rotations. Because of this, the structure of the tree changes continuously according to usage patterns.
- ✓ A splay tree gives amortized O(log n) time complexity for operations. This means that although one operation may sometimes be slow, the average performance of many operations is efficient.

❖ Operations in Splay Tree

1. Insertion Insertion is first done like a normal BST insertion. After inserting the new node, rotations are applied to move the inserted node to the root of the tree.

2. Deletion

- ✓ To delete a node:
- ✓ First search the node.
- ✓ If it has no child, remove it directly.
- ✓ If it has one child, replace it with its child.
- ✓ If it has two children, find the successor, swap values, and delete the succes

3. Searching Searching is performed like a BST search. If the element is found, it is moved to the root using rotations. If not found, the last visited node becomes the root.

❖ Rotations in Splay Tree

- ➢ **Zig Rotation** Similar to a single right rotation in AVL trees. Nodes move one step to the right.
- ➢ **Zag Rotation** Similar to a single left rotation in AVL trees. Nodes move one step to the left.
- ➢ **Zig-Zig Rotation** A double right rotation where nodes move two positions to the right.
- ➢ **Zag-Zag Rotation** A double left rotation where nodes move two positions to the left.
- ➢ **Zig-Zag Rotation** Combination of zig then zag rotation. Node moves right then left
- ➢ **Zag-Zig Rotation** Combination of zag then zig rotation. Node moves left then right

Advantages of Splay Tree

- ✓ Simple structure
- ✓ No need for balance factors or colors
- ✓ Frequently used nodes become faster
- ✓ Good average performance

Disadvantages of Splay Tree

- ✓ A single operation may take O(n)
- ✓ Tree may become temporarily unbalanced
- ✓ Performance is not always predictable
- ✓ Not suitable for strict real-time systems

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